



FRAUD DETECTION IN FINANCIAL TRANSACTIONS

Areej Alsulami and Rana Alabdan

Department of Computer Sciences
College of Computer and Information Sciences
Majmaah University
Al-Majmaah, 11952, Saudi Arabia
e-mail: 441204477@s.mu.edu.sa

Department of Information Systems
College of Computer and Information Sciences
Majmaah University
Al-Majmaah, 11952, Saudi Arabia
e-mail: r.alabdan@mu.edu.sa

Abstract

Incidents of fraud are increasing intensely considering the growth of modern technology. Globally, the increase has led to the loss of billions of dollars annually. Though, the use of preventive technologies remains the most effective measure to reduce fraud, it is crucial to understand that fraudsters are vastly flexible and are ready to explore methods to escape such measures. Using machine learning, we

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emphasis on an advanced and innovative fraud detection system. The study uses historical transaction statistics in training predictive models to recognize fraudulent transactions and lessen incorrect positives correctly. This study focuses on aiding the reduction in financial fraud and improving the security and dependability of digital financial transactions by merging data preprocessing, feature engineering, and model evaluation. The paper uses machine learning models such as logistic regression, CatBoost, decision trees (DTs), and random forests to compare performances and pick the best one for the study case. Accuracy, *F*-scores, precision, recall, and support values determine model performance. The results in this paper indicate that DT and CatBoost provide the highest level of accuracy, at 99.9% and 99%, respectively. These discoveries establish the potential of machine learning in significantly improving fraud detection, thereby curbing financial losses and improving the security of digital transactions.

1. Introduction

The evolution of businesses and individuals has amplified online fraud in the digital era with banks spending over \$70 billion annually on compliance costs. Global economy has been bedeviled by Cybercrimes with a cost of about \$600 billion yearly, with credit card fraud leading the line. This shocking inclination highlights the urgent need for potent fraud detection methods. Financial fraud has increased lately with the use of technological advancements like the internet and mobile computing. Fraud detection is vital to safeguard the menace of fraud defend the integrity and security of economic systems and protect individuals, businesses, and financial institutions from probable losses. Fraud detection purposely uses unlawful methods or practices to obtain financial gain [1]. It has a substantial negative influence on business, society, and everything else. It has bigger implications for the industry, like providing funds for illegal activities, drug trafficking and organized crime. The main goal of fraud detection is to differentiate between genuine and fraudulent transactions by leveraging advanced techniques and technologies. Anomaly detection, machine learning algorithms, behavioral analysis, and real-time monitoring are an array of methodologies fraud detection approaches. Historical

statistics/data used to build predictive models that classify transactions as fraudulent or genuine based on learned patterns are analyzed by machine learning algorithms. Real-time monitoring systems continuously analyze incoming transaction data, promptly raising alerts for possible fraudulent transactions [2]. Fraudsters are incessantly growing their techniques, and there is a necessity for detection methods to evolve accordingly. The failure of fraud prevention has brought fraud detection into play, and the limited exchange of ideas in fraud detection has made the development more difficult. Benford's law, which has existed since the late 18th century, is the simplest way to detect fraud. Machine learning is capable of analyzing millions of data points to detect efforts at fraud that would take a huge amount of time for humans to complete. Machine learning also actively learns and calibrates to improve accuracy and decrease false rejections. It can also function on massive datasets that humans cannot instantly access. Machine learning approaches fall into two essential categories: supervised and unsupervised. Supervised training contains anomalies constantly to be identified as before, and lots of supervised methods have been used over the last few decades to detect credit card fraud. Highly imbalanced databases seem to remain the major impediment in executing machine learning for detecting fraud [3].

2. Types of Financial Fraud

There are numerous types of financial frauds, such as credit card fraud, insurance fraud, securities and commodities fraud, money laundering, mortgage fraud, and lots of others. A brief explanation of some of the major types of frauds is listed below.

2.1. Insurance fraud

Insurance fraud can be well-defined as abusing an insurance policy to gain unlawful benefits from an insurance business [4]. Usually, the responsibility of insurance is to protect organizations or individuals' transactions against any financial risk [5]. The main sectors besieged by fraudulent insurance claims include healthcare [6] and automobile insurance

companies. Insurance fraud can ensue at several points in the insurance process (e.g., application, eligibility, rating, billing, and claims). Customers, brokers, agents, and staff members of insurance companies, healthcare providers, and others can commit to it [7]. A common form of claimed fraud is automobile insurance fraud, often committed by staging or intentionally committing accidents that result in excessive repair and injury costs. Insurance fraud can also include extreme billing, duplicate claims, and kickbacks to brokers [8].

2.2. Securities and commodities fraud

Securities fraud, or commodities fraud, refers to various methods by which false information is used to deceive a person into investing in a company. These include Pyramid Schemes, Market Manipulation, The Ponzi Scheme, Hedge Fund Fraud, Foreign Exchange Fraud, and Broker Embezzlement [8].

2.3. Money laundering

It is the process by which criminals hide or mask the earnings of their crimes or translate those earnings into goods and services. It permits criminals to inject their illicit funds into the stream of commerce, demeaning financial institutions and the money supply and unjustifiably empowering criminals with economic powers [7]. Similarly, money laundering is defined as the process by which criminals “wash dirty money” hide its unlawful source, and make it appear genuine and clean [9].

2.4. Credit card fraud

Credit card fraud is known as the unauthorized use of a person's credit card to implement fraudulent transactions without the user's knowledge [8]. A stolen or lost physical card can be implemented for transactions, but it is often executed remotely [10]. The cardholder's information may be accessed by one of a few methods. The phishing method involves a fraudster impersonating a finance official to convince the card user to reveal their details, and swipers or skimmers provide a front to an ATM or POS device that can read the card directly. A distinctive method for detecting credit card

fraud is to analyze a customer's expenditure patterns and flag transactions that are noticeably and outrageously outside this model [1]. The two major credit card frauds are application and behavioral fraud [11].

2.5. Mortgage fraud

It is a specific form of financial fraud that involves manipulating property or mortgage documents. It is often committed to changing the value of a property to encourage a lender to finance a loan for it [8]. Figure 1 represents a few financial fraud categories that transactions can fall into.

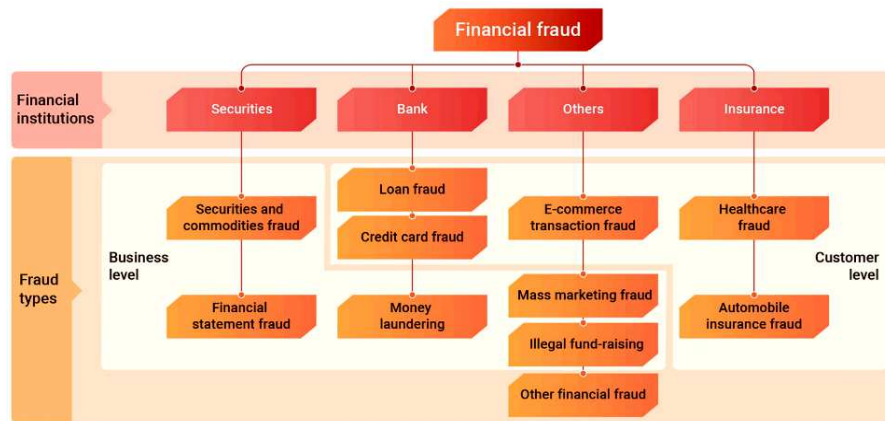


Figure 1. Common types of financial fraud. Taken from [12].

3. Literature Review

Researchers across the globe have contributed to emerging fraud detection forecasts using ML algorithms. A critical analysis of the current state-of-the-art of research in this field is provided by the following literature review.

The hidden Markov model was discussed which may aid the discovery of fraudulent transactions and include different transaction ranges such as low, medium, and high as the remark symbols. The hidden Markov model made the fraud detection system quite easy taking less time despite its complex process. Ideally, the hidden Markov model requires training in

annotated data. In addition, this model also needs manual markup. So, this model is inappropriate for detecting unknown patterns and is unusable in recognizing new fraud patterns [13].

Alghofaili et al. [14] proposed a profound learning-based method for detecting financial fraud using the Long Short-Term Memory (LSTM) technique. This model aims to augment detection techniques and accuracy in the light of huge data. A real dataset of credit card frauds is utilized to evaluate the suggested model, and the outcomes are compared with an existing deep learning model named the auto-encoder model and some other machine learning techniques. The results generated from the experiment demonstrated a flawless performance of LSTM, achieving 99.95% accuracy in less than a minute.

Karpoff [15] noted that the growth in platforms involving crowd funding is most possible to facilitate more fraudulent transactions and cases. However, when viewing the overall trends in fraud cases, different factors are considered. Some areas where blockchain is integrated further into their transactions show decreased fraud cases. It also highlights the massive work needed to detect and decrease fraud generally. Given the multifaceted nature of finance and the involved transactions, the predictions might be difficult to pull off.

Scams like credit card fraud have soared with the expansion of e-banking and other online payment options. As an alternative to using a straightforward voting process, it has been proposed utilizing an ensemble model based on sequential data modeling using deep recurrent neural networks and a novel voting mechanism based on artificial neural networks to uncover fraudulent actions. The experimental results on two real-world datasets show that the suggested model performs better than the most recent models [16].

Mytnyk et al. [17] offered the recognition of banking fraud using artificial intelligence algorithms. It also specifies several methods for advancing detection accuracy in financial domains. Based on an artificial

neural network, the proposed model proficiently enhances the accuracy of fraudulent transaction detection. The outcomes of the numerous algorithms are visualized, and the best results come from the logistic regression algorithm, which has an AUC value of approximately 0.946. The results show that the optimal AUC values for the decision tree, naive Bayes, *K*-nearest neighbor, and SVC algorithm classifiers are 0.938., 0.908, 0.927, and 0.936, respectively.

Zhou et al. [18] concluded that most fraud-detection systems use at least one supervised learning method and unsupervised and semi-supervised learning methods. The study showed that these techniques may be employed in combination to build more vigorous classifiers or in unitary. These methods are proportional and successful in detecting fraud and credit grading. They emphasized that classification-related problems, such as feature engineering, parameter selection, and hyperparameter tuning, apply to fraud detection and data-mining-based credit grading.

Carcillo et al. [19] predicted fraudulent transactions using three machine learning (ML) algorithms: logistic regression, naive Bayes, and *K*-nearest neighbor. Its comparative study records the performance of these algorithms. The model for predicting fraud based on logistic decline was superior to other prediction models.

Gupta et al. [20] tried to device an automated model using different machine learning techniques for the detection of these kinds of frauds; the projected model applied four algorithms used in machine learning, namely, naive Bayes, random forest, logistic regression, and SVM, on a vast dataset to predict the fraud. Naïve Bayes algorithm performance is outstanding for detecting credit card fraud among all the ML algorithms according to the results, with an accuracy of 80.4% and an area under the curve of 96.3%.

Chung and Lee [21] suggested an enhanced algorithm to detect credit card fraud with high sensitivity. The method applied three machine learning models: *K*-nearest neighbors, linear discriminant analysis, and linear regression. Additional conditional statements, such as “if” and “then,”

were next applied, and triggers like “>” and “<” to the results. Using this proposed strategy, the features were extracted and accomplished a recall of 1.0000, 0.9701, 1.0000, and 0.9362 across the four tested fraud datasets. Accordingly, this method has outclassed other methods using single machine learning recall models.

Awoyemi et al. [22] examined the performance of the data from *K*-nearest neighbor, logistic regression, and naive Bayes applied to credit card transaction data, which was further resampled using Synthetic Minority Over-sampling Technique (SMOTE). The result revealed that the *K*-nearest neighbor performed better than the logistic regression and naive Bayes measured regarding recall, precision, balanced classification rate, specificity, and Mathew’s correlation coefficient. The results show optimal accuracy for naive Bayes, *K*-nearest neighbor, and logistic regression classifiers as 97.92%, 97.69%, and 54.86%, respectively.

4. Methodology

This work utilizes classification algorithms, including LR, RF, DT, and CatBoost. The outcome of each method will be compared to identify the best strategy. The methodology adopted in this study to classify non-fraudulent transactions from fraudulent transactions is discussed in this section, including data collection and pre-processing using a machine learning algorithm. This section explicates dataset selection and uses machine learning models such as logistic regression, CatBoost, decision tree, and random forest to give room for the comparison of performances and the selection of the optimal for the study case. The method of decisive performance will be the accuracy values, *F*-scores, precision, recall, and support values for the models.

4.1. Dataset

The dataset for this research, called “fraudulent transactions,” was gathered in 2017 and obtained from an open-source resource on Kaggle [23]. The dataset contains 6,362,620 records with 11 features. 1,467,967 is the

mean value of all the transactions. There were 6,362,620 transactions, with 8,213 fraudulent transactions and 6,354,407 non-fraudulent ones. However, the dataset is highly imbalanced. There is an intense data imbalance where 0.1% of the transactions are fraudulent; this problem will be resolved in the next section. In this dataset, the focus feature is fraud, a binary feature with 0 (not fraud) and 1 (fraud). There are 6,354,407 non-fraudulent transactions (99.87%) and 8,213 fraudulent transactions (0.13%). The features of the fraudulent transaction dataset are shown in Table 1.

Table 1. Variables and explanations

Features	Description
Step	Maps a unit of time in the real world. In this case 1 step is 1 hour of time. Total steps 744 (30 days simulation)
Type	Type of transaction: CASH-IN, CASH-OUT, DEBIT, PAYMENT and TRANSFER
Amount	Amount of the transaction in local currency
nameOrig	Customer who started the transaction
oldbalanceOrg	Initial balance before the transaction
newbalanceOrig	New balance after the transaction
nameDest	Customer who is the recipient of the transaction
oldbalanceDest	Initial balance recipient before the transaction
newbalanceDest	New balance recipient after the transaction
isFraud	Determines if transaction is fraudulent (encoded as 1) or valid (encoded as 0)
isFlaggedFraud	Determines if transaction is flagged as fraudulent (encoded as 1) or not flagged at all (encoded as 0). An observation is flagged if the transaction is fraudulent, and it involved a transfer of over 200,000 in the local currency

Table 2. Dataset description using Python programming

	step	type	amount	nameOrig	oldbalanceOrg	newbalanceOrig	nameDest	oldbalanceDest	newbalanceDest	isFraud	isFlaggedFraud
0	1	2	9839.64	C1231006815	170136.0	160296.36	M1979787155	0.0	0.0	0	0
1	1	2	1864.28	C1666544295	21249.0	19384.72	M2044282225	0.0	0.0	0	0
2	1	3	181.00	C1305486145	181.0	0.00	C553264065	0.0	0.0	1	0
3	1	1	181.00	C840083671	181.0	0.00	C38997010	21182.0	0.0	1	0
4	1	2	11668.14	C2048537720	41554.0	29885.86	M1230701703	0.0	0.0	0	0

4.2. Exploratory data analysis

4.2.1. Data cleaning

Data cleaning, also known as data cleansing or scrubbing, discovers and eliminates mistakes and discrepancies from data to improve data quality. Data quality problems are present in a single data collection, such as files and databases, e.g., due to misspellings during data entry, missing information, or other invalid data. Data cleansing involves recognizing and correcting errors, contradictions, and duplicate entries in a dataset to improve data quality. As shown in Figure 2, no null values exist in the dataset.

```
Missing data:
step          0
type          0
amount        0
nameOrig      0
oldbalanceOrg 0
newbalanceOrig 0
nameDest      0
oldbalanceDest 0
newbalanceDest 0
isFraud       0
isFlaggedFraud 0
dtype: int64
```

Figure 2. Data cleaning in the dataset.

4.3. Data visualization

In this subsection, we review some visualizations used in the dataset to illustrate the presence of certain features.

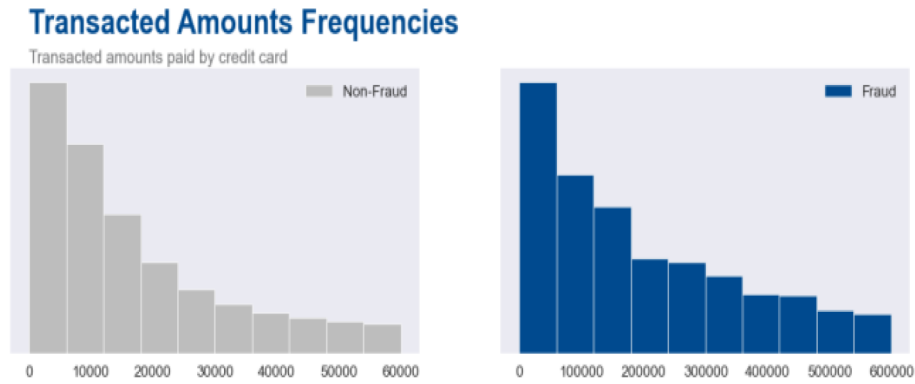


Figure 3. Transacted amounts paid by credit card.

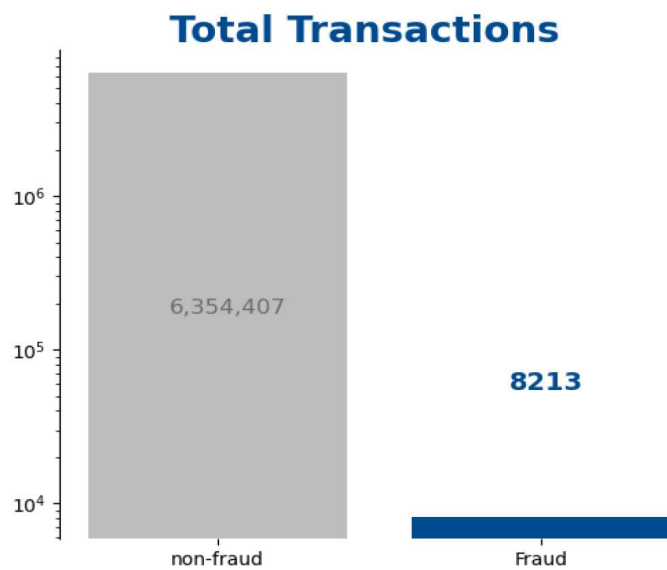


Figure 4. The total of transactions (fraud vs non-fraud case).

4.4. Statistics

In this subsection, an illustration of the relationship and strength of correlation among the features present in the dataset areas is shown in Figure 5.

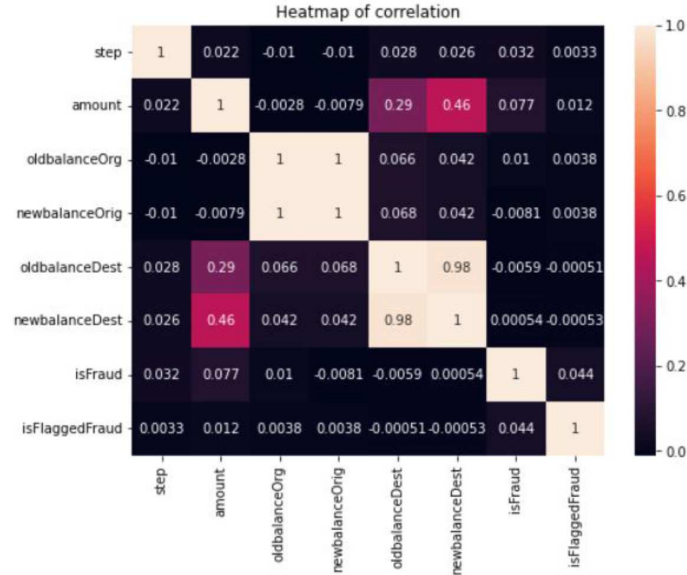


Figure 5. The correlation matrix between features.

5. Machine Learning (ML) Techniques

ML is a subfield of artificial intelligence (AI), it emphasizes on developing algorithms and models that allow computers to learn from data and make predictions or decisions without being explicitly programmed. It involves training computer systems to automatically examine and extract meaningful designs or intuitions from large datasets. This paper uses ML techniques such as LR, RF, DT, and CatBoost. A summary of each algorithm is now presented.

5.1. Logistic regression (LR)

LR techniques are commonly used in binary and multi-class classification problems [1]. Regression is performed on a set of variables. LR is largely valid for unfolding designs clarifying connections between numerous dependent binary variables. LR is one of the most used ML techniques for detecting financial misappropriations. It allows modeling a binary variable in terms of a set of projected random variables, whether

numerical or class-based. LR is used to envisage the likelihood of an incident happening with supplementary knowledge of the values of variables that can be interpreted or related to that occurrence [24].

5.2. Random forest (RF)

RF is an algorithm used for classification and regression. It builds a forest of individual decision trees that collectively constitute an ensemble. Each tree makes a proposal based on the data by taking popular votes. As a result, the class with the popular vote makes the final proposal. Thus, individual uncorrelated models align to provide the best proposal [25].

5.3. Decision trees (DTs)

DT is an ML technique for creating decision-support tools in trees of inner nodes instead of binary options [26]. Financial fraud has been detected via several DT-based methods for several years. A DT provides a graphical illustration of thinkable solutions to a choice based on situations. A DT begins with a root node, which is divided into discrete branches, then connected with other nodes, and so on. DTs end up in a node called a *leaf node*. Each node in a DT denotes a test; its associated branches denote its possible results, and a leaf node is assigned a class label.

5.4. CatBoost

CatBoost is a machine learning algorithm that boosts gradients in decision trees. It handles huge datasets and solves intricate problems swiftly and accurately. It uses gradient boosting and features to distinguish data types, making it perfect for categorization and ranking. CatBoost is robust to outliers, can learn from imbalanced datasets, and requires three minutes to train [27].

5.5. Model accuracy

We now compare the accuracy of the above algorithms and determine which algorithm attains the highest precision.

Table 3. Comparison of the results achieved by the algorithms tested

ML models		Precision	Recall	F1-score	Accuracy
DT	0	1.00	1.00	1.00	99.9%
	1	0.89	0.87	0.88	
LR	0	0.91	0.94	0.92	92%
	1	0.94	0.90	0.92	
RF	0	1.00	0.98	0.99	98%
	1	0.059	0.98	0.11	
CatBoost	0	1.00	0.99	1.00	99%
	1	0.63	0.85	0.72	

According to the above results, DT models produce better results. The algorithms used in this study are more accurate, cost-effective, and faster. In addition, our DT and CatBoost models achieve 99.9% and 99% accuracy, respectively. The inclusion of more qualities in our dataset improves the accuracy level. Additionally, our results indicate that RF outperforms the LR classifier in detecting fraud. The results indicate that DT delivers a more accurate prediction of fraud in the early stages than other methods.

6. Results and Discussions

While most researchers use SVC and DT to detect fraud, these results demonstrate that RF and LR models produce better results [28]. We apply more accurate, cost-effective, and faster algorithms in the present study. In conclusion, the decision tree is superior to the logistic regression algorithm regarding recall, AUC, and confusion matrix. Furthermore, there will always be room for improvement even with the good results achieved by the models. Further performance comparisons can be carried out by using other classification models and parameter optimization specific to each algorithm.

7. Conclusion

Financial fraud has developed predominantly in recent years. Stealing personal data, identities, funds, and sensitive information, posing a significant risk to individuals and institutions has risen among individuals and organizations. Fraudsters can now use electronic communication

channels and the internet to carry out sophisticated and advanced fraudulent activities with technological advancements. Using machine-learning techniques, we conducted comprehensive research to detect financial fraud. Massive datasets of financial data were analyzed, and algorithms were trained to recognize potential fraud patterns. The results establish the selected algorithms' efficiency and effectiveness in fraud detection. These algorithms rely on studying the financial data of individuals and institutions, including purchasing patterns, financial activities, and unusual transactions. Using machine learning techniques, projective models are built to identify and predict financial fraud behaviors. In this paper, several methods are included. Classification algorithms like LR, RF, DT, and CatBoost have been examined. The results indicate that DT and CatBoost provide the highest level of accuracy, at 99.9% and 99%, respectively. We object to keep working on financial datasets where the dataset varies over time, and we need to retrain.

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